Seongtae Bang

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TECHNICAL INTERESTS

GPU & Heterogeneous Systems, AI Training Systems, GPU Memory Management, Runtime & Parallel Execution, Computer Architecture

EDUCATION

Daegu Gyeongbuk Institute of Science and Technology (DGIST) **Mar. 2020 - Feb. 2027 (Expected)**
Ph.D. Student, Electrical Engineering and Computer Science

Daegu Gyeongbuk Institute of Science and Technology (DGIST) **Mar. 2016 - Feb. 2020**
B.E., School of Undergraduate Studies

SKILLS & EXPERTISE

Programming: C, C++, CUDA C/C++, Python, OpenMP, AVX-512 SIMD

AI Frameworks & Training: PyTorch, DeepSpeed (ZeRO-3), Megatron-LM, gsplat, mixture-of-experts, distributed training

GPU & CPU-GPU Systems: NVIDIA open-source UVM driver, GPU memory management, CPU-GPU communication/overlap, NUMA

Profiling & Simulation: Nsight Systems/Compute, GPGPU-Sim, Accel-Sim, gem5, dist-gem5

OPEN-SOURCE CONTRIBUTIONS

DeepSpeed — ZeRO-3 CPU Offloading and Large-Scale Training **2026**

- Identified a ZeRO-3 CPUAdam correctness issue in repeated same-step subgroup updates, developed a fix and regression tests, and **merged the fix upstream** (PR #7982).
- Submitted ReFlow for upstream integration as an asynchronous CPU-offloaded ZeRO-3 implementation, including optimizer kernels, gradient handling, and correctness tests (PR #8719).
- Reviewed CPUAdam fusion work (PR #8841) and shared additional optimization directions for the standard CPU-offload path through the DeepSpeed Q4 roadmap (#8672).

EXPERIENCES

Graduate Researcher, CASLAB **Mar. 2020 - Feb. 2027**
Yonsei University, Seoul (Sep. 2024 - Feb. 2027) · **DGIST**, Daegu (Mar. 2020 - Aug. 2024)
Conducted Ph.D. research with Prof. Daehoon Kim across GPU memory systems, heterogeneous execution, and AI training; continued at Yonsei after the lab moved, while remaining enrolled at DGIST.

Teaching Assistant — Computer Architecture, DGIST **Mar. 2021 - Jul. 2021**
Undergraduate Computer Architecture course.

Industry Intern — Winitech Co., Ltd. **Dec. 2017 - Feb. 2018**
Company-University Cooperation (CUop) Program.

FGLP Participant — University of California, Berkeley **Jun. 2016 - Aug. 2016**
Freshmen Global Leadership Program (FGLP).

RESEARCH ACTIVITIES

ReFlow / ReplayOpt: CPU-Offloaded LLM Training **ASPLOS 2027 / IEEE CAL 2026**

- Conventional offloading approaches primarily hide host-side update latency; profiling instead identified **unnecessary host-memory traffic and repeated gradient movement** as the dominant costs, rather than CPU optimizer arithmetic.
- Redesigned DeepSpeed ZeRO-3 around CPU-side gradient processing, register-resident SIMD fusion, and dependency-aware Ready/Persist scheduling, separating GPU-critical parameter generation from optimizer-state work that can finish later.
- Validated training semantics and regression behavior across Adam/AdamW and Lion, clipping, overflow, checkpointing, and single-/multi-GPU systems including **NVIDIA B200, A100, and RTX PRO GPUs**; achieved up to **4.0× throughput** over ZeRO-Infinity in evaluated configurations.

Code: https://github.com/st-bang97/DeepSpeed/tree/PR_branch
Example: https://github.com/st-bang97/DeepSpeedExamples/tree/PR_branch

ReclaimX: Device-Side Reclamation under UVM Oversubscription **MICRO 2026**

- Rather than accepting the host-driven UVM reclamation path while GPU resources wait on page faults, reinterpreted **fault-stalled GPU execution capacity** as an opportunity to perform reclamation.
- Designed fine-grained victim ordering, selective compressed residency, and GPU-local restoration to reduce host-mediated recovery; evaluated the architecture through cycle-level GPU/UVM simulation.

Code: https://github.com/st-bang97/ReclaimX_MICRO2026

RESEARCH ACTIVITIES (CONTINUED)

NVIDIA UVM Driver Research — ARIADNE / SAFE
HPCA 2026 / IEEE CAL 2025

- Conventional UVM migration and prefetching heuristics can move data that does not match application sharing or access patterns; analyzed page faults, runtime sharing, and evolving memory demand to identify ineffective transfers.
- Modified NVIDIA open-source UVM driver paths to adapt memory-management and prefetch decisions to observed workload behavior.

Code: https://github.com/st-bang97/ARIADNE_HPCA26

pNet-gem5: Full-System Network Simulation
IEEE CAL 2025

- Identified missing multi-queue NIC and guest-network support in dist-gem5-based full-system simulation, limiting the ability to model parallel packet processing.
- Implemented a **multi-queue gem5 NIC model and corresponding guest driver**, integrating the modeled device with the OS network stack.

Code: https://github.com/caslab-yonsei/pNet-gem5

Memory-Scalable Mixture-of-Experts Training
Under Review · Megatron-LM

- Existing offloading choices trade HBM capacity against efficient grouped expert execution; analyzed routing demand and grouped-GEMM granularity to redesign parameter residency and update timing.
- Profiled and modified Megatron-LM CPU optimizer and training paths to reduce host-side data movement while keeping routed expert computation grouped and GPU-efficient.

Code: Planned release after peer review; Megatron-LM upstream contribution planned.

Memory-Scalable 3D Gaussian Splatting Training
Under Review · gsplat

- Moving large 3DGS state off GPU memory leaves GPU-side culling residency or creates a new CPU culling bottleneck; instead, redesigned the culling algorithm for efficient CPU execution.
- Overlapped future-view CPU culling and preparation with current-view GPU rasterization/backpropagation, reducing model-proportional GPU residency without delaying the GPU pipeline.

Code: Planned release after peer review.

HONORS & FUNDING

MICRO 2026 Student Travel Grant — **2026**
59th IEEE/ACM International Symposium on Microarchitecture

Government Scholarship (Graduate), DGIST — **Mar. 2020 - Feb. 2025**

Faculty Council Scholarship, DGIST — **Mar. 2019 - Feb. 2020**

Government Scholarship (Undergraduate), DGIST — **Mar. 2016 - Feb. 2020**

PUBLICATIONS

[ASPLOS 2027] Seongtae Bang, Gyeongseo Park, Ki-Dong Kang, Hyunkyun Shin, Sungju Kim, and Daehoon Kim. **ReFlow: Exposing Parallelism in CPU-Offloaded LLM Training via Register-Resident SIMD Fusion and Decoupled Update Scheduling.** ACM International Conference on Architectural Support for Programming Languages and Operating Systems, 2027.

[MICRO 2026] Seongtae Bang, Hyunkyun Shin, Hyungwon Park, Minho Kim, and Daehoon Kim. **ReclaimX: Device-Side Memory Reclamation via Stalled GPU Execution for UVM Oversubscription.** IEEE/ACM International Symposium on Microarchitecture, 2026.

[HPCA 2026] Hyunkyun Shin, **Seongtae Bang**, Hyungwon Park, and Daehoon Kim. **ARIADNE: Adaptive UVM Management for Efficient GPU Memory Oversubscription.** IEEE International Symposium on High-Performance Computer Architecture, 2026.

[IEEE CAL 2026] Seongtae Bang, Gyeongseo Park, Kyeonghyeon Ryu, and Daehoon Kim. **ReplayOpt: Optimizer-State Replay to Resolve Critical-Path Bottlenecks in Offloaded Training.** IEEE Computer Architecture Letters, 2026.

[IEEE CAL 2025] Hyunkyun Shin†, **Seongtae Bang†**, Hyungwon Park, and Daehoon Kim. **SAFE: Sharing-Aware Prefetching for Efficient GPU Memory Management with Unified Virtual Memory.** IEEE Computer Architecture Letters, 2025. † Equal contribution.

[IEEE CAL 2025] Jongmin Shin, **Seongtae Bang**, Gyeongseo Park, and Daehoon Kim. **pNet-gem5: Full-System Simulation with High-Performance Networking Enabled by Parallel Network Packet Processing.** IEEE Computer Architecture Letters, 2025.